

Question 1 : Joins

Write an SQL query to generate the following output by comparing two tables, SOURCE and TARGET, using joins and conditional logic (Medium)(KOCH)

1. Detect records that are in the SOURCE table but not in the TARGET table and label them as "New in Source".
2. Detect records that are in the TARGET table but not in the SOURCE table and label them as "New in Target".
3. Detect records that are in both tables but have different NAME values and label them as "Mismatch".

The query should combine results from both tables, including appropriate labels, and order the output by ID. The output should look like this. Write the SQL query to achieve this output.

ID Comment

3 New in Source

4 Mismatch

5 New in Target

Given the following data in SOURCE and TARGET tables:

SOURCE Table:

ID NAME

1 A

2 B

3 C

4 D

TARGET Table:

ID NAM

1 A

2 B

4 X

5 E

-- Assignment Solutions

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create database interview_questions;

use interview_questions;

-- Create SOURCE Table

CREATE TABLE SOURCE (

ID INT PRIMARY KEY,

NAME VARCHAR(10)

);

-- Create TARGET Table

CREATE TABLE TARGET (

ID INT PRIMARY KEY,

NAME VARCHAR(10)

);

-- Insert Data into SOURCE Table

INSERT INTO SOURCE (ID, NAME) VALUES (1, 'A');

INSERT INTO SOURCE (ID, NAME) VALUES (2, 'B');

INSERT INTO SOURCE (ID, NAME) VALUES (3, 'C');

INSERT INTO SOURCE (ID, NAME) VALUES (4, 'D');

-- Insert Data into TARGET Table

INSERT INTO TARGET (ID, NAME) VALUES (1, 'A');

INSERT INTO TARGET (ID, NAME) VALUES (2, 'B');

INSERT INTO TARGET (ID, NAME) VALUES (4, 'X');

INSERT INTO TARGET (ID, NAME) VALUES (5, 'E');

-- 3 , New in source

-- 5, New in target

-- 4, Mismatch

-- join the 2 tables

-- will find new in source

```
select s.id, 'New in Source' as Comment
from
source s
left join target t
ON s.id = t.id
WHERE t.id is NULL
UNION ALL
```

-- will find new in target

```
select t.id, 'New in Target' as Comment
from
source s
right join target t
ON s.id = t.id
WHERE s.id is NULL
UNION ALL
```

-- will find the mismatch

```
select s.id, 'Mismatch' as Comment
from
source s
inner join target t
ON s.id = t.id
WHERE s.name != t.name;
```

=====

-- will find new in source

```
select s.id,
CASE
WHEN t.id is NULL Then 'New in Source'
WHEN s.name != t.name Then 'Mismatch'
END AS Comment
from
source s
```

left join target t

ON s.id = t.id

WHERE t.id is NULL OR s.name!=t.name

UNION ALL

-- will find new in target

select t.id, 'New in Target' as Comment

from

source s

right join target t

ON s.id = t.id

WHERE s.id is NULL